Assignment-8: Deterministic Skip List

Problem Statement

You are required to implement a **Skip List** in C++ using **templates**, supporting the following operations:

- insert
- delete
- search
- print

The skip list will store only **keys (no values)** in sorted order.

Unlike a traditional skip list, node levels are **not generated randomly**. Instead, they are determined using a **predefined sequence of coin flips from input**, ensuring deterministic behavior.

Input Format

1. Key Type

```
<key_type>  //Allowed values:int, float
```

Example:

```
int
```

2. Maximum Level

```
<max_level>
```

Example:

```
5
```

3. Coin Flip Sequence

A sequence of 0s and 1s:

- 1 → promote to next level
- 0 → stop

```
<coin_flip_sequence>
```

Example:

```
1 1 0 0 1 0 1 1 0
```

For the sequence 1 1 0 0 1 0 1 1 0, the first insertion uses 1 1 0 giving level 3, the second uses 0 giving level 1, the third uses 1 0 giving level 2, and the fourth uses 1 1 0 giving level 3. Thus, each insertion consumes part of the sequence in order and determines the level accordingly.

4. Number of Queries

`<n>`

5. Operations

```
insert <key>
delete <key>
search <key>
print
```

Skip List Rules

Node Level Construction

- Start at level 1. For each 1 of coin flip sequence, increase level. Stop at first 0 or when max level is reached

Deterministic Behavior

- No randomness allowed. Use the given coin sequence strictly in order

Structure Requirements

- Store only keys (no values). Maintain sorted order. Use forward pointers. Maintain a head node

Template Requirement

```cpp
template <typename T>
class SkipList {
    // implementation
};
```

Output Format

Insert

```
Inserted
```

Delete

```
Deleted or Not Found
```

Search

```
Found or Not Found
```

Print

Print level-wise from top to bottom, do not print the empty levels.

```
Level 4: key key key
Level 3: ...
Level 2: ...
Level 1: ...
```

Example

Input

```
int
4
1 1 0 0 1 0 1 1 0
7
insert 10
insert 20
insert 15
print
search 15
delete 20
print
```

Output

```
Inserted
Inserted
Inserted
Level 3: 10
Level 2: 10 15
Level 1: 10 15 20
Found
Deleted
Level 3: 10
Level 2: 10 15
Level 1: 10 15
```

Constraints

- Number of operations $\leq 10^4$
- Maximum level ≤ 20
- Keys are unique